ARXA - Prediction has value through a feasible action

Private portfolio design study. Not a client deployment.

Problem and decision user

An industrial analytics initiative can optimize a prediction metric without improving the operational decision or economics. The proposed user is operations manager, asset owner and finance sponsor.

Operating and economic approach

A model for managing data-enabled operational transformation and investment choices. The package joins the business question, product scope, operating handoffs, technology/data assumptions and financial consequences. Existing tools and simpler alternatives remain part of the decision.

Checked result

Eight tests and twelve efficacy/capacity scenarios show how intervention assumptions can reverse a favorable cost comparison. The act/defer calculation is a cost comparison, not a real-option price.

Capabilities illustrated

Industrial analytics; action constraints; economic scenarios; staged investment. These are demonstrated documentary/model mechanisms, not a certification of personal independent mastery.

Limits and next use

Distribution shift, sensor quality and intervention effectiveness may invalidate estimated savings. Use the detailed specification and validation memo to scope a real mandate, identify missing data and define the smallest decisive test. Publication and deployment require separate review.

Delivery specification

Private design study. The first product is a decision package for one maintenance intervention, not a general industrial AI platform. The economic question is whether an executable action improves expected operating cost after intervention, capacity, data and integration costs.

User, value and alternatives

Proposed users: asset/operations owner, maintenance planner, data steward and finance reviewer. A forecast matters only if an authorized operator can take a timely, safe action. The adoption hypothesis is an internal decision-support pilot; no factory deployment, paid demand, causal maintenance effect or realized saving is established.

Compare no intervention, a simple fixed rule, forecast-triggered action and a cost/capacity-constrained policy. A more accurate score is not automatically a better decision. Prefer an existing maintenance-management process where the additional data/model cost exceeds the incremental benefit.

Data and model contract

Required real inputs: asset identity, timestamp/grain, sensor units/calibration, maintenance history, failure definition, exposure time, missingness, downtime cost, action cost, lead time, resources and safety constraints. Preserve temporal ordering and distinguish observations before and after intervention. Do not train or evaluate on future information.

The reference generates 80 deterministic periods. Its risk score is an engineered index, not an empirically calibrated failure probability. The rolling forecast predicts that index; failure labels are generated by a separate deterministic rule. Forecast MAE therefore does not measure calibrated failure risk. This fixture is a mechanism demonstration and must not support an investment return claim.

For new data, reconcile labels and units, establish a temporal holdout, define drift/missingness controls and compare with a simple baseline before estimating business value. Unknown intervention effectiveness remains an uncertainty scenario, not 100% by default in a real decision.

Operating process and ownership

1. Operations owner selects one asset, failure mode and decision window; finance owner records material costs and acceptable exposure.
2. Data steward checks timestamps, units, missing records and measurement changes. Unsafe or unexplained data pauses the model-assisted recommendation.
3. Analyst forecasts only from available past observations and documents uncertainty. Model owner compares action policies using the same external cases.
4. Maintenance planner checks crew, spare parts, lead time, production windows and capacity. A predicted failure without an executable intervention is an escalation, not a successful model outcome.
5. Authorized operator decides; safety constraints override the score. Log action, reason and observed result separately from forecast.
6. Reviewer compares predicted and observed outcomes, including false alarms, missed failures, intervention failures and opportunity cost. New evidence changes the policy or pauses it.

Economics and financing

Separate the historical technical fixture's large synthetic cost units from the management case in EUR per period. The management case assumes failure probability 0.10, failure loss 1000, two interventions at 30 and platform cost 20: expected cost 180. At failure probability 0.30 it is 380. This does not estimate the intervention's causal effect; it only shows sensitivity to assumed residual risk.

The technical policy originally assumed perfect prevention at every triggered action. The extended test includes efficacy 0, 0.5 and 1 and capacity limits. When efficacy is zero, intervention expense can worsen total cost. Capacity zero recovers the no-intervention baseline. No future outcome is used to prioritize actions in this reference; capacity is consumed chronologically.

The legacy function named real_option is now explicitly a one-step assumed act/defer cost comparison. Its cost gap is not an option price. A proper staged-investment valuation would require states, information arrival, transition assumptions, admissible actions, discounting and financing constraints. These remain design options linked to Arctervix, not implemented findings.

Architecture, delivery and risk

Layers: approved observations -> quality checks -> time-aware forecast -> action feasibility -> economic scenario -> human decision -> outcome log. Keep a separate source/assumption ledger and controlled dashboard. Reuse maintenance systems and data tooling before building a platform; adapters and hosted services require separate access and confidentiality review.

Dashboard: data completeness, forecast error with sample size, candidate versus feasible actions, capacity used, action cost, residual failure exposure and assumed total cost. Distinguish predicted index, calibrated probability and actual event. No single aggregate score should conceal a safety constraint or unavailable spare part.

Material risks: leakage, weak labels, distribution shift, false confidence in efficacy, unavailable intervention resources and overlapping economic benefits. Owners are respectively model/data reviewer, operations specialist, model owner, maintenance owner, planner and finance reviewer. Stop a policy if it fails the same-case cost comparison or violates safety; retain the underlying technical knowledge for a better-scoped option.

Handoff and narrative

Deliver the data card, temporal evaluation plan, policy assumptions, scenarios, capacity/efficacy results, limitations and decision memo. The next real pilot requires an approved dataset and a defined operational decision, not mandatory user coursework.

Interview narrative: translate predictive analytics into an executable operating choice. Review exposed perfect-prevention assumptions and an overstated real-options label. The corrected package demonstrates how effectiveness and capacity can reverse the economic conclusion. It is a documented model review, not evidence of industrial savings or production deployment.

Validation evidence and reflection

Private synthetic study. Eight repository tests; twelve efficacy/capacity scenarios.

Efficacy	Capacity	Cost	Actions	Residual failures
0	0	540000	0	6
0	2	584000	2	6
0	10	760000	10	6
0	unlimited	1024000	22	6
0.5	0	540000	0	6
0.5	2	584000	2	6.0
0.5	10	670000	10	5.0
0.5	unlimited	754000	22	3.0
1	0	540000	0	6
1	2	584000	2	6
1	10	580000	10	4
1	unlimited	484000	22	0

No-intervention baseline cost: 540000. Synthetic technical cost units; not the EUR/period management case. Rolling index MAE: 0.039256; degraded-input MAE: 0.039728. This predicts an engineered index, not calibrated failure probability.

Decision and forecast interpretation

The original favorable-policy forecast holds only under its selected perfect-efficacy assumptions. Zero efficacy reverses the advantage; zero capacity reproduces no intervention. Preserve the negative result instead of presenting only the favorable scenario. The separate pre-recorded management-cost example is 180 versus 380 EUR/period at residual failure probabilities 0.10 and 0.30. This arithmetic does not identify treatment benefit. Continue a narrowly scoped decision study. Pause an operating recommendation until data quality, failure labels, efficacy and available intervention resources are established. A staged capital commitment requires a separate cash-flow and financing model.

Post-mortem

Removed the misleading option_value field: an absolute cost difference is not a real-option price. Added efficacy/capacity scenarios and rejected prediction-length mismatch instead of silently truncating observations. Costs, scores and probabilities retain distinct meanings. Handoff: DELIVERY-SPECIFICATION.md, tests and artifacts/management-review.json. No real asset, customer, industrial saving or production integration is claimed.